MAXIMUM TWO-DISTANCE SETS IN EUCLIDEAN SPACES OF DIMENSION UP TO TWENTY-ONE

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Abstract. A two-distance set is a finite set of points in $\mathbb{R}^d$ whose pairwise distances take exactly two values; $g(d)$ denotes the largest cardinality of such a set. The values $g(d)$ were known for $d \le 8$ (Einhorn–Schoenberg, Lisoněk) and, since a recent computer-assisted classification, for $9 \le d \le 12$, where $g(d) = \binom{d+1}{2}$ is attained by the midpoints of the edges of a regular simplex. We prove that $g(d) = \binom{d+1}{2}$ for every d with $10 \le d \le 21$, and that the simplex-midpoint set $\mathcal{J}(d+1,2)$ is the only two-distance set of that size up to similarity. The underlying statement, valid in every dimension $d \ge 10$, is that a set in $\mathbb{R}^d$ whose squared distances are 2 and 4 has at most $\binom{d+1}{2}$ points, with equality only for $\mathcal{J}(d+1,2)$. The proof replaces the exhaustive clique searches that were used in dimensions 9 to 12 by a combinatorial argument. Each component of such a set under its short distances is, up to translation, a subset of a root lattice; in the lattice A_n it becomes a uniform family of subsets with two intersection sizes, so that the Ray-Chaudhuri–Wilson bound $\binom{n+1}{2}$ applies and the Ahlswede–Khachatrian theorem identifies the extremal families; in D_n it becomes a binary code with distances 2 and 4, for which the bound $1 + \binom{n}{2}$ of Barg and Musin applies; and root lattices with several irreducible components are handled by a slicing lemma whose only numerical inputs are a few invariants of E_6, E_7 and E_8, computed by a small clique search. For $13 \le d \le 21$ the remaining distance ratios allowed by the Larman–Rogers–Seidel theorem are excluded by the recent bounds of Chen and Yu. All computations are reproduced by a second, independently written program, and the method recovers the previously known values $g(10)$, $g(11)$, $g(12)$ and the bound 45 for the root lattices of rank 9.

1. Introduction

A finite set $X \subset \mathbb{R}^d$ is a *two-distance set* if the set $\{\|x-y\| : x \ne y \in X\}$ has exactly two elements. Let $g(d)$ be the largest cardinality of a two-distance set in $\mathbb{R}^d$. The problem of determining $g(d)$ goes back to Kelly [15] and Croft [9]; the general bounds

$$\binom{d+1}{2} \le g(d) \le \binom{d+2}{2} \tag{1}$$

are due to the midpoint construction and to Blokhuis [6], improving Bannai, Bannai and Stanton [2]. The lower bound is attained by the set

$$\mathcal{J}(m,2) = \{e_i + e_j : 1 \le i < j \le m\} \subset \mathbb{R}^m, \qquad m = d+1,$$

of midpoints of the edges of a regular simplex (scaled), which lies in the hyperplane $x_1+\cdots+x_m = 2$ and has squared distances 2 and 4. Einhorn and Schoenberg [12] settled $d \le 3$, Lisoněk [17] classified the maximal two-distance sets for $d \le 7$ by computer and found a 45-point set in $\mathbb{R}^8$ attaining Blokhuis' bound, so that

$$g(1), \dots, g(8) = 3, 5, 6, 10, 16, 27, 29, 45.$$

For $d \ge 9$ the problem remained open until recently. In [21] it was shown, by an exhaustive symmetry-reduced clique search in graphs attached to the root lattices of rank $9, 10, 11, 12$, that $g(9) = 45$ (attained by $\mathcal{J}(10,2)$ and by Lisoněk's set), and that $g(d) = \binom{d+1}{2}$ for $d = 10, 11, 12$,

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attained only by $\mathcal{J}(d+1,2)$. The concluding section of [21] points out that, thanks to the bounds of Chen and Yu [7] on two-distance sets with other distance ratios, the determination of $g(d)$ for all $d \le 21$ is reduced to computing the clique numbers of the same graphs for root lattices of rank up to 21, but that the graphs grow so quickly ($G(D_{21})$ has 96,642 vertices) that "new ideas will be needed well before $d = 21$". The present paper supplies these ideas and completes the program.

Theorem 1.1. *Let $d \ge 10$ and let $X \subset \mathbb{R}^d$ be a finite set all of whose pairwise squared distances lie in $\{2,4\}$. Then $|X| \le \binom{d+1}{2}$, with equality if and only if X is congruent to $\mathcal{J}(d+1,2)$.*

Theorem 1.2. *For every d with $10 \le d \le 21$ one has $g(d) = \binom{d+1}{2}$, and every two-distance set of cardinality $g(d)$ in $\mathbb{R}^d$ is similar to $\mathcal{J}(d+1,2)$.*

Together with [21], Theorem 1.2 determines $g(d)$ for all $d \le 21$; the only dimension in $\{9,\dots,21\}$ with a second extremal configuration is $d = 9$. Dimensions 22 and 23 are genuinely different: the McLaughlin graph gives a spherical two-distance set of 275 points in $\mathbb{R}^{22}$ [18, 5], and Ge, Koolen and Munemasa [13] constructed a two-distance set of 277 points in $\mathbb{R}^{23}$; both have squared-distance ratio $2/3$ rather than $1/2$, so that they are not covered by Theorem 1.1, and the bounds of [7] do not exclude them. See Section 9.

The route from Theorem 1.1 to Theorem 1.2 is the one of [21]. By the theorem of Larman, Rogers and Seidel [16] in Neumaier's form [19], a two-distance set with more than $2d+1$ points has squared distances in the ratio $(k-1)/k$ for an integer k with $2 \le k \le \frac{1}{2} + \sqrt{d/2}$; for $d \le 12$ only $k = 2$ is possible, and for $13 \le d \le 23$ also $k = 3$. Sets with $k = 2$ are, after scaling, exactly the sets of Theorem 1.1. Sets with $k = 3$ in $\mathbb{R}^d$ have at most $24(d+1)/(24-d)+1$ points by [7, Theorem 3.1], which is smaller than $\binom{d+1}{2}$ for $d \le 21$ (and larger for $d = 22, 23$).

The proof of Theorem 1.1 starts, as in [21], from the observation that the integer vectors $x - y$, $x, y \in X$, generate an even lattice and that the short differences (of norm 2) generate a root lattice R; if the graph of short distances is connected then X is, up to translation, a subset of R, and in any case each connected component of that graph is. Thus everything reduces to the following purely combinatorial quantity: for a root lattice R, let

$$\mathcal{N}(R) = \max\{|X| : X \subset R,\ 0 \in X,\ \|x-y\|^2 \in \{2,4\} \text{ for all distinct } x, y \in X\},$$

so that $\mathcal{N}(R) = 1 + \omega(G(R))$ in the notation of [21], where $G(R)$ is the graph on the vectors of norm 2 and 4 of R in which two vectors are adjacent when their difference has norm 2 or 4. In [21] the numbers $\mathcal{N}(R)$ were computed for all 566 root lattices of rank at most 12. Here we determine them, or bound them sharply enough, for all root lattices. The three ingredients are as follows.

The lattices A_n. A set $X \subset A_n \subset \mathbb{Z}^{n+1}$ with squared distances 2 and 4 has the property that every coordinate takes at most two values on X (Lemma 3.1). Hence, after a translation, X consists of 0/1-vectors of a fixed weight w, that is, X is a family of w-subsets of an $(n+1)$-set any two of which intersect in $w-1$ or $w-2$ elements. The theorem of Ray-Chaudhuri and Wilson [22] gives $|X| \le \binom{n+1}{2}$, and the complete intersection theorem of Ahlswede and Khachatrian [1] shows that the families with at least $\binom{n+1}{2} - 1$ members are $\mathcal{J}(n+1,2)$ and $\mathcal{J}(n+1,2)$ minus a point (Theorem 3.2). This replaces the clique searches in $G(A_n)$, the largest graphs of [21], by a two-line argument valid for all n.

The lattices D_n. Here two coordinates may differ by 2, but only along a single "diametral" pair of points in each such coordinate direction (Lemma 4.1); removing the diametral pairs leaves a set of 0/1-vectors with Hamming distances 2 and 4, that is, a binary code with two distances, for which Barg and Musin [4] proved the bound $1 + \binom{n}{2}$. We obtain $\mathcal{N}(D_n) = 1 + \binom{n}{2}$ for all $n \ge 7$ (Theorem 4.2).

Reducible root lattices. For $R = R' \oplus R''$ we slice a set $X \subset R$ along the second coordinate. The slices, and the set of second coordinates, are again sets with squared distances 2 and 4, and points in different slices are strongly constrained. This yields an upper bound for $\mathcal{N}(R' \oplus R'')$ in terms of $\mathcal{N}(R')$, $\mathcal{N}(R'')$ and two further invariants of R' and R'': the largest number $a(R)$ of roots pairwise at angle 60°, and the largest size $\nu_k(R)$ of a set of roots with pairwise inner products 0 or 1 that contains k roots at inner product 1 with all the others (Lemma 5.1). For A_n and D_n these invariants are computed by hand; for E_6, E_7, E_8 they are computed by a clique search in graphs with at most 240 vertices, which we carry out with two independent programs.

Putting the three ingredients together we prove (Theorem 6.1) that for $d \geq 10$ every root lattice R of rank at most d other than A_d satisfies $\mathcal{N}(R) \leq \binom{d+1}{2} - 2$; for $10 \leq d \leq 22$ this is checked by evaluating the bound of Lemma 5.1 for each root lattice (there are 5,713 of rank 22), and for $d \geq 15$ it also follows from explicit estimates, so that the statement holds in every dimension. The geometric Theorem 1.1 then follows by the reduction of [21], which we recall with proofs in Section 7.

As a by-product we obtain the exact values $\mathcal{N}(A_n) = \binom{n+1}{2}$ and $\mathcal{N}(D_n) = 1 + \binom{n}{2}$ $(n \geq 7)$ in all ranks, an answer in the range $10 \leq d \leq 22$ to the question raised in [21, Remark 8.3] about the largest configurations not coming from A_d (they come from D_d and have $1 + \binom{d}{2}$ points, Corollary 6.2), and an independent confirmation of the values $g(10)$, $g(11)$, $g(12)$ of [21] by a different method (for $d = 9$, Theorem 6.1 recovers the bound 45 for the root lattices of rank 9, which is the computational core of [21] in that dimension). The computations are small: apart from the verification runs of Section 8, the entire proof needs only the clique searches for the invariants of E_6, E_7, E_8, which take seconds.

2. Preliminaries

Throughout, $\langle \cdot, \cdot \rangle$ is the standard inner product and $\|x\|^2 = \langle x, x \rangle$. For $m \geq 1$ we write $[m] = \{1, \dots, m\}$ and $e_1, \dots, e_m$ for the standard basis.

2.1. **Admissible sets.** A finite set X in a Euclidean space is a $\{2,4\}$-*set* if $\|x - y\|^2 \in \{2, 4\}$ for all distinct $x, y \in X$. (We allow sets in which only one of the two values occurs; a $\{2,4\}$-set with more than $d+1$ points in $\mathbb{R}^d$ uses both values.) By Blokhuis' theorem [6],

$$|X| \leq \binom{d+2}{2} \qquad \text{for every } \{2,4\}\text{-set } X \subset \mathbb{R}^d. \tag{2}$$

For a lattice L we call $X \subset L$ *admissible* if $0 \in X$ and X is a $\{2,4\}$-set, and we put

$$\mathcal{N}(L) = \max\{|X| : X \subset L \text{ admissible}\}.$$

Since $X - p$ is admissible for every $\{2,4\}$-set $X \subset L$ and every $p \in X$, $\mathcal{N}(L)$ is simply the largest size of a $\{2,4\}$-subset of L. If $L \subset \mathbb{R}^r$ then $\mathcal{N}(L) \leq \binom{r+2}{2}$ by (2).

2.2. **Root lattices.** An *even lattice* is a lattice $\Lambda \subset \mathbb{R}^r$ with $\|\lambda\|^2 \in 2\mathbb{Z}$ for all $\lambda \in \Lambda$; then $\langle \lambda, \mu \rangle \in \mathbb{Z}$ for all $\lambda, \mu \in \Lambda$. A *root lattice* is an even lattice generated by its vectors of norm 2, its *roots*. By Witt's classification (see [11, Ch. 1] or [8, Ch. 4]) every root lattice is an orthogonal direct sum of irreducible root lattices of types A_n $(n \geq 1)$, D_n $(n \geq 4)$, E_6, E_7, E_8, uniquely up to order. We use the standard coordinates

$$A_n = \Big\{x \in \mathbb{Z}^{n+1} : \sum x_i = 0\Big\}, \qquad D_n = \Big\{x \in \mathbb{Z}^n : \sum x_i \equiv 0 \pmod 2\Big\},$$
$$E_8 = D_8 \cup \big(D_8 + \tfrac{1}{2}(1, \dots, 1)\big),$$

and E_7, E_6 are the sublattices of E_8 orthogonal to a root, respectively to an A_2-sublattice. For a root lattice R let R_2 and R_4 be the sets of vectors of norm 2 and 4. The roots of A_n are the

vectors $e_i - e_j$, its norm-4 vectors are $e_i + e_j - e_k - e_l$ with i, j, k, l distinct; the roots of D_n are $\pm e_i \pm e_j$, its norm-4 vectors are $\pm 2e_i$ and $\pm e_i \pm e_j \pm e_k \pm e_l$. In a direct sum $R' \oplus R''$ the roots are the roots of the summands and the norm-4 vectors are those of the summands together with all sums $r' + r''$ of a root of R' and a root of R''. The Weyl group $W(R)$ is generated by the reflections in the roots; it preserves R, R_2, R_4 and all distances, and permutes the irreducible components of R of the same type.

The following two invariants of a root lattice R enter the bound for direct sums.

Definition 2.1. Let R be a root lattice. A set $Y \subset R_2$ is *admissible* if $\langle y, y' \rangle \in \{0, 1\}$ for all distinct $y, y' \in Y$ (equivalently, $\{0\} \cup Y$ is an admissible set). A set $K \subset R_2$ is a $60°$-*clique* if $\langle y, y' \rangle = 1$ for all distinct $y, y' \in K$. Put

$$\begin{aligned} a(R) &= \max\{|K| : K \subset R_2 \text{ a } 60°\text{-clique}\}, \\ \nu_0(R) &= \max\{|Y| : Y \subset R_2 \text{ admissible}\}, \\ \nu_k(R) &= \max\{|Y| : Y \subset R_2 \text{ admissible}, \ Y \supset K \text{ for a } 60°\text{-clique } K \text{ with } |K| = k \\ &\qquad \text{and } \langle y, w \rangle = 1 \text{ for all } w \in K, \ y \in Y \setminus K\} \end{aligned}$$

for $1 \le k \le a(R)$, and $\nu_k(R) = 0$ for $k > a(R)$. We also write $\nu_+(R) = \max_{k \ge 1} \nu_k(R)$.

Thus $1 + \nu_0(R)$ is the largest size of an admissible set consisting of 0 and roots, and $\nu_k(R)$ concerns admissible sets of roots with k "universal" roots at inner product 1 with everything else. Clearly $a(R) \le \operatorname{rank} R$ (the Gram matrix of a $60°$-clique is $I + J$, which is nonsingular), $\nu_k(R) \le \nu_1(R) \le \nu_0(R)$ for $k \ge 1$, and $\nu_k(R) \ge k$ for $k \le a(R)$.

Lemma 2.2. *For root lattices R', R'': $a(R' \oplus R'') = \max\{a(R'), a(R'')\}$, $\nu_0(R' \oplus R'') = \nu_0(R') + \nu_0(R'')$, and $\nu_k(R' \oplus R'') = \max\{\nu_k(R'), \nu_k(R'')\}$ for $k \ge 1$.*

Proof. Roots of $R' \oplus R''$ lie in R' or in R'', and roots in different summands have inner product 0. Hence a $60°$-clique lies in one summand, an admissible set is the union of admissible sets in the two summands, and in a set counted by ν_k with $k \ge 1$ the $60°$-clique K lies in one summand, say R', and every other member has inner product 1 with the members of K, hence also lies in R'. □

The values of the invariants for the irreducible root lattices are determined in Sections 3, 4 and 5 (Table 1).

2.3. **Set systems.** We use two classical theorems. A family $\mathcal{F}$ of w-subsets of $[m]$ is L-*intersecting* if $|A \cap B| \in L$ for all distinct $A, B \in \mathcal{F}$, and t-*intersecting* if $|A \cap B| \ge t$ for all $A, B \in \mathcal{F}$.

Theorem 2.3 (Ray-Chaudhuri–Wilson [22])**.** *If $\mathcal{F}$ is a family of w-subsets of $[m]$ that is L-intersecting for a set L of s non-negative integers, then $|\mathcal{F}| \le \binom{m}{s}$.*

Theorem 2.4 (Ahlswede–Khachatrian [1])**.** *Let $1 \le t \le w \le m$ with $m \ge 2w - t$. The largest size of a t-intersecting family of w-subsets of $[m]$ is $\max_{i \ge 0} |\mathcal{F}_i|$, where $\mathcal{F}_i = \{A \in \binom{[m]}{w} : |A \cap [t+2i]| \ge t + i\}$.*

For binary codes we use the following result of Barg and Musin ([4, Theorem 12(a)]; see also [3, Theorem 1.1] and [10, Theorem 1.4] for the exact value in the case of interest).

Theorem 2.5 (Barg–Musin)**.** *Let $C \subset \{0, 1\}^m$ be a code in which exactly two Hamming distances $d_1 < d_2$ occur between distinct codewords. If $d_1 + d_2 \le m$, then $|C| \le 1 + \binom{m}{2}$.*

We write $M(m)$ for the largest size of a subset of $\{0,1\}^m$ all of whose pairwise Hamming distances lie in $\{2,4\}$. A code with a single distance 2 or 4 is the vertex set of a regular simplex of side $\sqrt{2}$ or 2 in $\mathbb{R}^m$ and has at most $m+1$ points; together with Theorem 2.5 this gives

$$M(m) = 1 + \binom{m}{2} \quad (m \geq 6), \qquad M(0), \dots, M(5) = 1, 1, 2, 4, 8, 16. \tag{3}$$

Indeed $\{0\} \cup \{e_i + e_j : i < j\}$ has distances 2 and 4; all codewords of a $\{2,4\}$-code have weights of the same parity, which gives $M(m) \leq 2^{m-1}$, attained for $m \leq 5$ by the even-weight vectors (for $m \leq 5$ these have pairwise distances 2 or 4), and $M(1) = 1$, $M(0) = 1$ are trivial.

3. The lattices A_n

Lemma 3.1. *Let $X \subset A_n \subset \mathbb{Z}^{n+1}$ be a $\{2,4\}$-set. Then for every coordinate $i \in [n+1]$ the set $\{x_i : x \in X\}$ has at most two elements, and they are consecutive integers.*

Proof. For $x \neq y$ in X the vector $x - y$ is a root $e_j - e_k$ or a norm-4 vector $e_j + e_k - e_l - e_m$ of A_n, and all these vectors have coordinates in $\{-1, 0, 1\}$. Hence $|x_i - y_i| \leq 1$ for all $x, y \in X$. □

Fix an admissible set $X \subset A_n$ and let $c_i \in \{-1, 0\}$ be the smaller of the (at most two) values of the i-th coordinate on X (it is 0 or -1 because $0 \in X$). Then $x = c + \mathbf{1}_{B(x)}$ with $B(x) = \{i : x_i = c_i + 1\}$, and since all $x \in X$ have coordinate sum 0, the sets $B(x)$ have the common size $w = -\sum_i c_i$. Moreover $\|x - y\|^2 = |B(x) \triangle B(y)| = 2\big(w - |B(x) \cap B(y)|\big)$. Consequently

$$X \longmapsto \mathcal{F}(X) = \{B(x) : x \in X\} \tag{4}$$

maps every admissible set $X \subset A_n$ injectively onto a family of w-subsets of $[n+1]$, $0 \leq w \leq n+1$, in which any two distinct members intersect in $w-1$ or $w-2$ elements, and $X = c + \{\mathbf{1}_B : B \in \mathcal{F}(X)\}$ is determined by $\mathcal{F}(X)$ up to a translation. Conversely every such family $\mathcal{F}$ gives an admissible set, namely $\{\mathbf{1}_B - \mathbf{1}_{B_0} : B \in \mathcal{F}\}$ for any $B_0 \in \mathcal{F}$. The family $\binom{[m]}{2}$ of all 2-subsets corresponds to $\mathcal{J}(m,2)$, and complementation $B \mapsto [m] \setminus B$ corresponds to the lattice automorphism $x \mapsto -x$ (up to translation).

Theorem 3.2. *$\mathcal{N}(A_n) = \binom{n+1}{2}$ for every $n \geq 2$, and $\mathcal{N}(A_1) = 2$. Moreover, for $n \geq 4$ every admissible set $X \subset A_n$ with $|X| \geq \binom{n+1}{2} - 1$ is, up to a translation and an automorphism of the lattice A_n (the group $\mathrm{Aut}(A_n) = W(A_n) \times \{\pm 1\}$), the set $\mathcal{J}(n+1,2)$ or $\mathcal{J}(n+1,2)$ with one point removed. In particular these sets are spherical, with circumradius ρ given by $\rho^2 = 2 - 4/(n+1)$.*

Proof. Let $m = n+1$ and let $\mathcal{F} = \mathcal{F}(X)$ be the family of w-subsets attached to an admissible X by (4). If $w \geq 2$ then $\mathcal{F}$ is L-intersecting with $L = \{w-1, w-2\}$, and Theorem 2.3 gives $|X| = |\mathcal{F}| \leq \binom{m}{2}$. If $w \leq 1$ then $\mathcal{F}$ has at most m members, and if $w \geq m-1$ at most m as well (a w-set with $w \geq m-1$ is determined by its complement, of size ≤ 1). Hence $\mathcal{N}(A_n) \leq \binom{n+1}{2}$ for $n \geq 2$, and $\mathcal{J}(n+1,2)$ shows equality. For $n = 1$ the lattice A_1 has no vectors of norm 4 and its two roots are at squared distance 8, so $\mathcal{N}(A_1) = 2$.

Now let $n \geq 4$, so $m \geq 5$, and $|\mathcal{F}| \geq \binom{m}{2} - 1$. As just seen, $w \in \{0, 1, m-1, m\}$ gives $|\mathcal{F}| \leq m < \binom{m}{2} - 1$. If $w = 2$ or $w = m-2$ then, after complementation if necessary, $\mathcal{F}$ is a subfamily of $\binom{[m]}{2}$ with at least $\binom{m}{2} - 1$ members, that is, all 2-subsets or all but one; as point sets these are $\mathcal{J}(m,2)$ and $\mathcal{J}(m,2)$ minus a point (any two points of $\mathcal{J}(m,2)$ are equivalent under the coordinate permutations $S_m = W(A_n)$). It remains to exclude $3 \leq w \leq m-3$. Such a family is t-intersecting with $t = w - 2 \geq 1$, and $m \geq w + 2 = 2w - t$, so by Theorem 2.4 its size

is at most $\max_i |\mathcal{F}_i|$ with

$$|\mathcal{F}_0| = \binom{m-w+2}{2}, \qquad |\mathcal{F}_1| = 1 + w(m-w), \qquad |\mathcal{F}_2| = \binom{w+2}{2}, \qquad \mathcal{F}_i = \emptyset \ (i \geq 3),$$

because $\mathcal{F}_i$ consists of the w-sets containing at least $w-2+i$ elements of a $(w-2+2i)$-set, which for $i \geq 3$ requires more than w elements. Since $3 \leq w \leq m-3$ we have $\binom{m-w+2}{2} \leq \binom{m-1}{2}$ and $\binom{w+2}{2} \leq \binom{m-1}{2}$, and $\binom{m-1}{2} = \binom{m}{2} - (m-1) < \binom{m}{2} - 1$; also $1 + w(m-w) \leq 1 + m^2/4 < \binom{m}{2} - 1$ as $m \geq 5$. So $|\mathcal{F}| < \binom{m}{2} - 1$, a contradiction.

The set $\mathcal{J}(m,2)$ lies on the sphere about $\frac{2}{m}(1,\dots,1)$ of squared radius $2 - 4/m$, and so does every subset. □

Remark 3.3. The extremal families for the Ray-Chaudhuri–Wilson bound with two intersection sizes are in general not unique; the point of the second half of the proof is that the specific intersection sizes $\{w-1, w-2\}$ force $w \in \{2, m-2\}$ once $|\mathcal{F}|$ is close to the bound. The values $|\mathcal{F}_i|$ can also be read off from [3, Theorem 4.1 and Table 1], where the maximum two-distance codes in the Johnson spaces $J^{m,w}$ are determined for several w; we only need the upper bounds.

Proposition 3.4. *For $n \geq 1$: $a(A_n) = n$, $\nu_0(A_n) = \lfloor (n+1)^2/4 \rfloor$ and $\nu_k(A_n) = n$ for $1 \leq k \leq n$.*

Proof. Two roots $e_i - e_j$ and $e_k - e_l$ of A_n have inner product $[i=k] + [j=l] - [i=l] - [j=k]$. An admissible set Y of roots therefore contains no two roots $e_i - e_j$, $e_k - e_i$ (their inner product is -1 or -2), so the sets P of "heads" i and M of "tails" j of the roots $e_i - e_j \in Y$ are disjoint and $Y \subset \{e_p - e_m : p \in P,\ m \in M\}$; conversely every such set is admissible, with all inner products in $\{0,1\}$. This gives $\nu_0(A_n) = \max_{|P|+|M| \leq n+1} |P||M| = \lfloor (n+1)^2/4 \rfloor$. Two roots have inner product 1 if and only if they share their head or their tail, so a 60°-clique is contained in $\{e_i - e_j : j \neq i\}$ or in $\{e_j - e_i : j \neq i\}$ for some i, and $a(A_n) = n$. Finally let Y be admissible and contain a root $w = e_1 - e_2$ with $\langle y, w \rangle = 1$ for all $y \in Y \setminus \{w\}$. Then $Y \setminus \{w\} \subset \{e_1 - e_m : m \neq 1,2\} \cup \{e_p - e_2 : p \neq 1,2\}$, and $\langle e_1 - e_m, e_p - e_2 \rangle = -[m=p]$, so the sets of indices m and p used are disjoint subsets of $[n+1] \setminus \{1,2\}$; hence $|Y| \leq 1 + (n-1) = n$. Thus $\nu_1(A_n) \leq n$, and $\nu_k(A_n) \leq \nu_1(A_n)$ for $k \geq 1$; the 60°-clique $\{e_1 - e_j : 2 \leq j \leq n+1\}$ shows $\nu_k(A_n) \geq n$ for all $1 \leq k \leq n$. □

4. The lattices D_n

Let $X \subset D_n \subset \mathbb{Z}^n$ be a $\{2,4\}$-set, $n \geq 4$. Now $x - y$ can be $\pm 2e_i$, and coordinates may take three values. Call a pair $\{x,y\} \subset X$ *diametral in direction i* if $y - x = \pm 2e_i$, and call i a *special* coordinate if such a pair exists.

Lemma 4.1. *Let $X \subset D_n$ be a $\{2,4\}$-set and let $\{x,y\}$ be diametral in direction i, say $y = x + 2e_i$. Then $z_i = x_i + 1$ for every $z \in X \setminus \{x,y\}$. Consequently each special direction contains exactly one diametral pair, different special directions have disjoint diametral pairs, and in a non-special coordinate the points of X take at most two consecutive values.*

Proof. Let $z \in X \setminus \{x,y\}$ and $t = z_i - x_i$, so that $(z-y)_i = t - 2$. Both $z - x$ and $z - y$ are vectors of norm 2 or 4 of D_n; the coordinates of such a vector lie in $\{-1,0,1\}$ unless the vector is $\pm 2e_j$ for some j, in which case its other coordinates vanish. Since t and $t-2$ both lie in $\{-2,\dots,2\}$, we have $t \in \{0,1,2\}$. If $t = 0$ then $z - y = -2e_i$, i.e. $z = x$; if $t = 2$ then $z - x = 2e_i$, i.e. $z = y$. Hence $t = 1$. If $\{x',y'\} \neq \{x,y\}$ were another diametral pair in direction i, it would contain a point $z \notin \{x,y\}$, with $z_i = x_i + 1$ by the first assertion; the other point $z \pm 2e_i$ of the pair has i-th coordinate $x_i + 3$ or $x_i - 1$, so it lies outside $\{x,y\}$ as well, contradicting the first assertion. If $\{x,y\}$ and $\{x,y'\}$ were diametral in directions $i \neq j$, then $\|y - y'\|^2 = \|2e_i - 2e_j\|^2 = 8$. The last assertion holds because in a non-special coordinate all differences are in $\{-1,0,1\}$. □

Theorem 4.2. *$\mathcal{N}(D_n) = 1+\binom{n}{2}$ for all $n \geq 7$. Moreover $\mathcal{N}(D_4) = 8$, $\mathcal{N}(D_5) = 16$, $\mathcal{N}(D_6) = 17$.*

Proof. The set $\{0\} \cup \{e_i + e_j : 1 \leq i < j \leq n\}$ shows $\mathcal{N}(D_n) \geq 1 + \binom{n}{2}$. For the upper bound let $X \subset D_n$ be a $\{2,4\}$-set, let S be the set of special coordinates, $s = |S|$, and let Z be the set of points of X that belong to no diametral pair, so that $|X| = |Z| + 2s$ by Lemma 4.1. By the lemma, all points of Z have the same coordinates in the special directions, and in each non-special coordinate they take two consecutive values. Hence Z is, up to a translation, a set of 0/1-vectors on the $n - s$ non-special coordinates with pairwise Hamming distances 2 and 4, and $|Z| \leq M(n - s)$ with M as in (3). Therefore

$$\mathcal{N}(D_n) \leq \max_{0 \leq s \leq n} \big(M(n - s) + 2s\big).$$

For $n - s \geq 6$ we have $M(n - s) + 2s = 1 + \binom{n-s}{2} + 2s \leq 1 + \binom{n}{2}$, because $\binom{n}{2} - \binom{n-s}{2} = \frac{s}{2}(2n - s - 1) \geq 2s$ for $s \leq 2n - 5$. For $n - s \leq 5$ we have $M(n - s) + 2s \leq 16 + 2s \leq 16 + 2n$, which is at most $1 + \binom{n}{2}$ for $n \geq 9$; for $n = 7$ and $n = 8$ one checks the finitely many values directly ($n = 7$: $M(5) + 4 = 20$, $M(4) + 6 = 14$, $M(3) + 8 = 12$, $M(2) + 10 = 12$, $M(1) + 12 = 13$, $M(0) + 14 = 15$, all below 22; $n = 8$: $16 + 6 = 22$, $8 + 8 = 16$, $4 + 10 = 14$, $2 + 12 = 14$, $1 + 14 = 15$, $1 + 16 = 17$, all below 29). This proves $\mathcal{N}(D_n) \leq 1 + \binom{n}{2}$ for $n \geq 7$.

For $n \leq 6$ the same bound gives $\mathcal{N}(D_4) \leq \max(M(4), M(3)+2, M(2)+4, M(1)+6, M(0)+8) = 9$, $\mathcal{N}(D_5) \leq \max(16, 10, 8, 8, 9, 11) = 16$ and $\mathcal{N}(D_6) \leq \max(16, 18, 12, 10, 10, 11, 13) = 18$, and the exact values 8, 16, 17 were obtained by a clique search in $G(D_n)$ (Section 8); the value 16 for D_5 is also in [21, Table 1]. A 17-point set in D_6 is $\{0\} \cup \{e_i + e_j : 1 \leq i < j \leq 5\} \cup \{e_1 + e_2 + e_3 + e_4 + e_5 - e_i : 2 \leq i \leq 5\} \cup \{e_1 \pm e_6\}$, that is, the diametral pair $e_1 \pm e_6$ together with the 15 vectors on the first five coordinates of weight 0 or 2, or of weight 4 and containing e_1. □

Remark 4.3. The values $\mathcal{N}(D_4) = 8$ and $\mathcal{N}(D_5) = 16$ are the even-weight halves of the cubes $\{0,1\}^4$ and $\{0,1\}^5$ (the 16-point set is the Clebsch configuration, the maximum two-distance set in $\mathbb{R}^5$), and $\mathcal{N}(D_6) = 17 > 1 + \binom{6}{2}$; the formula $1 + \binom{n}{2}$ starts at $n = 7$. For $n \geq 7$ the maximum is not unique: the vectors of weight 1 together with the weight-3 vectors containing a fixed coordinate also form a $\{2,4\}$-code of size $n + \binom{n-1}{2} = 1 + \binom{n}{2}$.

Proposition 4.4. *For $n \geq 4$: $a(D_n) = n-1$, $\nu_0(D_n) = \binom{n}{2}$ and $\nu_k(D_n) \leq 2n-3$ for $1 \leq k \leq n-1$, with equality for $k = 1, 2$.*

Proof. Two roots of D_n with disjoint supports have inner product 0; two roots $\epsilon e_i + \delta e_j$ and $\epsilon' e_i + \delta' e_k$ ($j \neq k$) sharing one index have inner product $\epsilon\epsilon'$; and two roots on the same pair $\{i, j\}$ have inner product ± 2 or 0, the latter when they agree in exactly one sign. A 60°-clique consists of roots whose supports pairwise share exactly one index with the same sign there; a family of 2-sets pairwise meeting in exactly one element is a star or a triangle, and the star $\{e_1 + e_j : 2 \leq j \leq n\}$ gives $a(D_n) = n - 1$ (a triangle has $3 \leq n - 1$ roots).

Let Y be an admissible set of roots. At most two roots of Y lie on a given pair $\{i, j\}$, and if two do, say $e_i + e_j$ and $e_i - e_j$ (after a sign change of coordinates, which is an automorphism of D_n), then no other root of Y involves j: a root $\pm e_j \pm e_k$, $k \neq i$, has inner product ± 1 with $e_i + e_j$ and the opposite value with $e_i - e_j$. Call j *blocked*; the pair $\{i, j\}$ uses i with a single sign, so i is not blocked, and distinct pairs carrying two roots block distinct indices. If t indices are blocked then $|Y| \leq 2t + \binom{n-t}{2}$, since the remaining roots of Y lie on distinct pairs of unblocked indices; and $2t + \binom{n-t}{2} \leq \binom{n}{2}$ because $\binom{n}{2} - \binom{n-t}{2} = \frac{t}{2}(2n - t - 1) \geq 2t$ for $t \leq 2n - 5$, which holds as $t \leq n - 1$ and $n \geq 4$. The set $\{e_i + e_j : i < j\}$ shows $\nu_0(D_n) = \binom{n}{2}$.

For ν_1 let Y be admissible and $w = e_1 + e_2 \in Y$ with $\langle y, w\rangle = 1$ for all other $y \in Y$. Then $Y \setminus \{w\} \subset \{e_1 \pm e_j, e_2 \pm e_j : 3 \leq j \leq n\}$. For a fixed $j \geq 3$, if both $e_1 + \epsilon e_j$ and $e_2 + \epsilon' e_j$ lie in Y

then $\epsilon = \epsilon'$ (otherwise the inner product is -1), and $e_1 + e_j$, $e_1 - e_j$ can both occur only if no root $e_2 \pm e_j$ does. So each $j \geq 3$ contributes at most two roots, and $|Y| \leq 1 + 2(n-2) = 2n-3$; the set $\{e_1 + e_2\} \cup \{e_1 \pm e_j : j \geq 3\}$ attains this, so $\nu_1(D_n) = 2n-3$, and $\nu_k \leq \nu_1$ for all $k \geq 1$. The admissible set $\{e_1 + e_2, e_1 + e_3, e_2 + e_3\} \cup \{e_1 \pm e_j : 4 \leq j \leq n\}$ has $2n-3$ elements and contains the 60°-clique $\{e_1 + e_2, e_1 + e_3\}$ of universal roots, so $\nu_2(D_n) = 2n-3$ as well. □

A computer enumeration (Section 8) gives the exact values $\nu_k(D_n) = 2n-k-2$ for $3 \leq k \leq n-1$ and $4 \leq n \leq 8$; we only use the bound $2n-3$.

5. Direct sums

Lemma 5.1 (Slicing lemma)**.** *Let $L = L' \oplus L''$ be an orthogonal direct sum of two even lattices and let $X \subset L$ be admissible. Then*

$$|X| \leq \max\Big\{\mathcal{N}(L'),\ \mathcal{N}(L''),\ \max_{\substack{1\leq u\leq a(L')+1\\ 1\leq s\leq a(L'')+1}} \big(\nu_{u-1}(L') + \nu_{s-1}(L'') + 1 + (u-1)(s-1)\big)\Big\},$$

where a and ν_k are defined for even lattices exactly as in Definition 2.1.

Proof. Write the points of X as $x = (x', x'')$ with $x' \in L'$, $x'' \in L''$. Let $U = \{x' : x \in X\} \subset L'$ and $S = \{x'' : x \in X\} \subset L''$; then $0 \in U \cap S$. For distinct $x, y \in X$ we have $\|x' - y'\|^2 + \|x'' - y''\|^2 = \|x - y\|^2 \in \{2, 4\}$, and both summands are non-negative even integers, since L' and L'' are even. Hence:

(a) if $x' \neq y'$ and $x'' \neq y''$, then $\|x' - y'\|^2 = \|x'' - y''\|^2 = 2$;
(b) U is a $\{2,4\}$-set in L' and S is a $\{2,4\}$-set in L'': for distinct $u, v \in U$ choose $x, y \in X$ with $x' = u$, $y' = v$; if $x'' = y''$ then $\|u - v\|^2 = \|x - y\|^2 \in \{2, 4\}$, and otherwise $\|u - v\|^2 = 2$ by (a).

For $u \in U$ let $F(u) = \{x'' : x \in X,\ x' = u\}$ be its fibre, and put $U_2 = \{u \in U : |F(u)| \geq 2\}$, $U_1 = U \setminus U_2$; define S_2, S_1 symmetrically. If $U_2 = \emptyset$ then $x \mapsto x'$ is injective on X, so $|X| = |U| \leq \mathcal{N}(L')$, because $U - u$ is admissible for any $u \in U$. Similarly $|X| \leq \mathcal{N}(L'')$ if $S_2 = \emptyset$. Assume now $U_2 \neq \emptyset \neq S_2$.

Claim: every $u_0 \in U_2$ satisfies $\|u_0 - v\|^2 = 2$ for all $v \in U \setminus \{u_0\}$. Indeed, let $(u_0, b), (u_0, b') \in X$ with $b \neq b'$ and let $(v, c) \in X$. Then $c \neq b$ or $c \neq b'$, say $c \neq b$, and (a) applied to (u_0, b) and (v, c) gives $\|u_0 - v\|^2 = 2$.

Consequently U_2 is a set of points pairwise at squared distance 2, and after the translation $U \mapsto U - u_0$ ($u_0 \in U_2$) the set U becomes $\{0\} \cup Y$ with $Y \subset L'$ a set of roots, pairwise at inner product 0 or 1 (as U is a $\{2,4\}$-set), containing the set $K = U_2 - u_0 \setminus \{0\}$ of $|U_2| - 1$ roots, which are pairwise at inner product 1 and have inner product 1 with every root of $Y \setminus K$ by the claim. Therefore

$$|U_2| - 1 \leq a(L'), \qquad |U| \leq 1 + \nu_{|U_2|-1}(L'),$$

and symmetrically $|S_2| - 1 \leq a(L'')$, $|S| \leq 1 + \nu_{|S_2|-1}(L'')$.

Now count X through the bipartite graph on $U \sqcup S$ whose edges are the points of X. Every $u \in U_1$ has degree 1, so the number of points of X with first coordinate in U_1 is $|U_1|$; the number of points with first coordinate in U_2 and second coordinate in S_1 is at most $|S_1|$ (each $b \in S_1$ has degree 1); and the number of points with first coordinate in U_2 and second coordinate in S_2 is at most $|U_2||S_2|$. Hence, with $u = |U_2|$ and $s = |S_2|$,

$$\begin{aligned} |X| &\leq |U_1| + |S_1| + us \leq \big(1 + \nu_{u-1}(L') - u\big) + \big(1 + \nu_{s-1}(L'') - s\big) + us \\ &= \nu_{u-1}(L') + \nu_{s-1}(L'') + 1 + (u-1)(s-1). \end{aligned}$$

□

Remark 5.2. The bound is attained in several situations. If $K' \subset L_2'$ and $K'' \subset L_2''$ are $60°$-cliques, then $(\{0\} \cup K') \times (\{0\} \cup K'')$ is admissible with $(|K'|+1)(|K''|+1)$ points (this is the term $u = a(L')+1$, $s = a(L'')+1$ with $\nu_a = a$); if $Y \subset L_2'$ is admissible and $K'' \subset L_2''$ is a $60°$-clique, then $(\{0\} \cup Y) \times \{0\} \cup \{0\} \times K''$ is admissible with $1 + \nu_0(L') + a(L'')$ points. For instance $\mathcal{N}(D_7 \oplus A_2) = 24 = 1 + \binom{7}{2} + 2$ and $\mathcal{N}(A_5 \oplus A_4) = 30 = 6 \cdot 5$ [21, Table 2]. For the 39 root lattices of rank 8 and the 57 root lattices of rank 9, the bound of Lemma 5.1 (minimised over the possible decompositions $L = L' \oplus L''$ and applied recursively, see (5)) is attained in 25, respectively 34, cases and never exceeds the true value by more than 10 (Section 8).

We now record the invariants of the exceptional lattices.

Proposition 5.3. *The invariants $a(R)$ and $\nu_k(R)$ of E_6, E_7, E_8 are as in Table 1. In particular $\nu_0(E_8) = 36$, $\nu_+(E_8) = 29$, $\nu_+(E_7) = 17$, $\nu_+(E_6) = 11$.*

R	$a(R)$	ν_0	ν_1	ν_2	ν_3	ν_4	ν_5	ν_6	ν_7	ν_8
A_n $(n \geq 1)$	n	$\lfloor (n+1)^2/4 \rfloor$	n	n	n	n	n	n	n	n
D_n $(n \geq 4)$	$n-1$	$\binom{n}{2}$	$2n-3$	$2n-3$	$\leq 2n-3$ ($= 2n-k-2$ for $n \leq 8$)					
E_6	5	16	11	11	7	5	5	–	–	–
E_7	7	27	17	17	11	8	7	7	7	–
E_8	8	36	29	29	19	14	11	9	8	8

TABLE 1. The invariants of the irreducible root lattices (ν_k for $k \leq a(R)$). The rows A_n, D_n are Propositions 3.4 and 4.4; the rows E_6, E_7, E_8 are computed (Section 8).

Proof. $\nu_0(R)$ is the clique number of the graph on R_2 in which two roots are adjacent when their inner product is 0 or 1, and $a(R)$ is the clique number of the graph on R_2 in which two roots are adjacent when their inner product is 1; these graphs have 72, 126, 240 vertices, and their clique numbers were computed with two independent programs (Section 8). For $k \geq 1$ we use that $W(R)$ acts transitively on R_2 and, for each root r, the stabiliser of r acts transitively on the roots at inner product 1 with r [8, Ch. 4]: for E_8 the stabiliser is $W(E_7)$ and these 56 roots are the minimal vectors of the dual of E_7, for E_7 the stabiliser is $W(D_6)$ acting on 32 half-spin vectors, and for E_6 it is $W(A_5)$ acting on the $20 = \binom{6}{3}$ roots $\frac{1}{2}(\pm 1, \dots)$ of the appropriate form. Hence every $60°$-clique K with $k \geq 2$ elements is $W(R)$-equivalent to one containing a fixed pair $\{r_1, r_2\}$ of roots at inner product 1, and every clique with $k = 1$ element to $\{r_1\}$. For $k \geq 1$, $\nu_k(R)$ is the maximum over the (finitely many) $(k-2)$-cliques K' in the common neighbourhood $N(r_1) \cap N(r_2)$ $(k \geq 2)$, respectively $K' = \emptyset$ $(k = 1)$, of $k + \omega_k$, where ω_k is the clique number of the graph on the roots at inner product 1 with all elements of $K = \{r_1, r_2\} \cup K'$ (resp. $\{r_1\}$), with adjacency "inner product 0 or 1". The common neighbourhood $N(r_1) \cap N(r_2)$ has 9, 15, 27 roots for E_6, E_7, E_8, so this is a small computation. The values were confirmed by a second program that enumerates all $60°$-cliques of R_2 (there are 1,501,680 nonempty ones in E_8) without using the transitivity (Section 8). □

Example 5.4. The value $\nu_0(E_8) = 36$ is attained by the 14 roots $e_1 \pm e_j$ $(2 \leq j \leq 8)$ together with the 22 roots $\frac{1}{2}(1, \epsilon_2, \dots, \epsilon_8)$ whose sign vector $(\epsilon_2, \dots, \epsilon_8)$ has exactly zero or two entries -1; the latter form a $\{2,4\}$-code of size $1 + \binom{7}{2} = 22$ in $\{0,1\}^7$, and $\frac{1}{2}(1, \epsilon) \cdot (e_1 \pm e_j) = \frac{1}{2}(1 \pm \epsilon_j) \in \{0,1\}$.

6. Admissible sets in root lattices of rank at most d

For a root lattice $R = R_1 \oplus \cdots \oplus R_m$ with irreducible components R_i of ranks r_i we define an upper bound $\overline{\mathcal{N}}(R)$ recursively. For $m = 1$ let $\overline{\mathcal{N}}(A_n) = \binom{n+1}{2}$ $(n \ge 2)$, $\overline{\mathcal{N}}(A_1) = 2$, $\overline{\mathcal{N}}(D_n) = 1 + \binom{n}{2}$ $(n \ge 7)$, which are the exact values of Theorems 3.2 and 4.2, and let $\overline{\mathcal{N}}(R) = \binom{r+2}{2}$ be the Blokhuis bound for $R = D_4, D_5, D_6, E_6, E_7, E_8$ of rank r. For $m \ge 2$ let

$$\overline{\mathcal{N}}(R) = \min_{R=R'\oplus R''} \max\Big\{\overline{\mathcal{N}}(R'), \overline{\mathcal{N}}(R''), \max_{u,s}\big(\nu_{u-1}(R') + \nu_{s-1}(R'') + 1 + (u-1)(s-1)\big)\Big\}, \tag{5}$$

the minimum over all decompositions of R into two nonempty sub-sums of components and the inner maximum over $1 \le u \le a(R') + 1$, $1 \le s \le a(R'') + 1$, where a and ν_k of a sub-sum are computed from Table 1 by Lemma 2.2 (for D_n we use $\nu_k(D_n) \le 2n - 3$, $k \ge 1$). By Lemma 5.1 and induction on m, $\mathcal{N}(R) \le \overline{\mathcal{N}}(R)$ for every root lattice R.

Theorem 6.1. *Let $d \ge 9$ and let R be a root lattice of rank at most d. Then $\mathcal{N}(R) \le \binom{d+1}{2}$. If $d \ge 10$ and $R \ncong A_d$, then $\mathcal{N}(R) \le \binom{d+1}{2} - 2$. If $d = 9$ and $\mathcal{N}(R) = 45$ then $R \cong A_9$ or $R \cong E_8 \oplus A_1$.*

Proof. We first dispose of small ranks. If $\operatorname{rank} R = r \le 8$ then $\mathcal{N}(R) \le \binom{r+2}{2} \le 45$ by (2). Let $r = 9$. If R is irreducible then $R = A_9$ with $\mathcal{N} = 45$ or $R = D_9$ with $\mathcal{N} = 37$. If R is reducible, the evaluation of (5) for the 55 reducible root lattices of rank 9 gives $\overline{\mathcal{N}}(R) \le 45$, with equality only for $E_8 \oplus A_1$ (the largest values are 45 for $E_8 \oplus A_1$, then 36 for $A_8 \oplus A_1$ and $E_7 \oplus A_2$); for $E_8 \oplus A_1$ the cross term in (5) is at most $\nu_0(E_8) + \nu_0(A_1) + 1 = 38$, so the bound 45 comes from $\overline{\mathcal{N}}(E_8) = 45$. This proves the statements for $d = 9$, and also

$$\mathcal{N}(R) \le \max\left\{\binom{r+1}{2}, 45\right\} \qquad \text{for every root lattice } R \text{ of rank } r \le 9. \tag{6}$$

Now let $d \ge 10$ and assume, by induction, that $\mathcal{N}(R) \le \binom{t+1}{2}$ for every root lattice of rank t with $10 \le t < d$; together with (6) this gives

$$\mathcal{N}(R) \le \max\left\{\binom{t+1}{2}, 45\right\} \le \binom{d}{2} \le \binom{d+1}{2} - 2 \qquad \text{if } \operatorname{rank} R = t \le d-1, \tag{7}$$

since $45 \le \binom{10}{2}$. So it suffices to consider R of rank exactly d. If R is irreducible then $R = A_d$ or $R = D_d$, with $\mathcal{N}(D_d) = 1 + \binom{d}{2} \le \binom{d+1}{2} - 2$. Let $R = R_1 \oplus R''$ be reducible, with R_1 irreducible of rank r and R'' of rank $t = d - r$, $1 \le r, t \le d - 1$. By Lemma 5.1 and (7), $\mathcal{N}(R) \le \max\{\binom{d+1}{2} - 2,\ T\}$ where

$$T = \max_{u,s}\big(\nu_{u-1}(R_1) + \nu_{s-1}(R'') + 1 + (u-1)(s-1)\big),$$

and we must show $T \le \binom{d+1}{2} - 2 = \binom{d}{2} + d - 2$. We use the following consequences of Table 1 and Lemma 2.2. For an irreducible root lattice L of rank ℓ,

$$\begin{gathered} \nu_0(L) \le \binom{\ell}{2} + \beta(L), \qquad a(L) \le \ell, \\ \nu_+(L) \le \varphi(\ell) := \max\{\ell,\ 2\ell - 3,\ 11[\ell \ge 6],\ 17[\ell \ge 7],\ 29[\ell \ge 8]\} \le 4\ell, \end{gathered} \tag{8}$$

where $\beta(A_1) = \beta(A_2) = \beta(A_3) = \beta(E_6) = 1$, $\beta(E_7) = 6$, $\beta(E_8) = 8$ and $\beta = 0$ otherwise; in all cases $0 \le \beta(L) \le \ell$ and $\beta(L) \le 8$. For $R'' = \bigoplus_{i\ge2} R_i$ with ranks r_i, $\sum_{i\ge2} r_i = t$, Lemma 2.2 gives $\nu_0(R'') \le \sum_{i\ge2}\big(\binom{r_i}{2} + r_i\big) \le \binom{t}{2} + t$ and $\nu_+(R'') \le \max_{i\ge2}\varphi(r_i) \le 4t$, $a(R'') \le t$. We distinguish the four cases according to whether $u = 1$ and $s = 1$.

Case $u = s = 1$. Then $T \le \nu_0(R_1) + \nu_0(R'') + 1 = \nu_0(R) + 1 \le \sum_{i=1}^{m} \binom{r_i}{2} + \sum_i \beta(R_i) + 1 \le \binom{d}{2} - \sum_{i<j} r_i r_j + d + 1$, using $\sum_i \binom{r_i}{2} = \binom{d}{2} - \sum_{i<j} r_i r_j$ and $\beta(R_i) \le r_i$. Since $m \ge 2$, $\sum_{i<j} r_i r_j \ge r_1(d - r_1) \ge d - 1 \ge 3$, so $T \le \binom{d}{2} + d - 2$.

Case $u \ge 2$, $s = 1$. Then $T \le \nu_+(R_1) + \binom{t}{2} + t + 1$, and $\binom{d}{2} - \binom{t}{2} = \frac{r}{2}(2d - r - 1)$, so $T \le \binom{d}{2} + d - 2$ as soon as $\nu_+(R_1) + 3 \le \frac{r}{2}(2d - r + 1)$. With the values $\nu_+(A_r) = r$, $\nu_+(D_r) \le 2r - 3$, $\nu_+(E_6) = 11$, $\nu_+(E_7) = 17$, $\nu_+(E_8) = 29$ of Table 1 this reads, respectively, $r + 1 + 6/r \le 2d$ (true since $r \le d - 1$ and $d \ge 7$), $r \le 2d - 3$, $d \ge 29/6$, $d \ge 41/7$ and $d \ge 15/2$. All hold for $d \ge 10$.

Case $u = 1$, $s \ge 2$. Then $T \le \nu_0(R_1) + \nu_+(R'') + 1 \le \binom{r}{2} + 8 + 4t + 1$, and $\binom{d}{2} - \binom{r}{2} = \frac{t}{2}(2d - t - 1) \ge \frac{td}{2}$ because $t \le d - 1$. So $T \le \binom{d}{2} + d - 2$ holds if $4t + 11 \le d + \frac{td}{2}$, i.e. if $11 \le d + t(d/2 - 4)$, which is true for $d \ge 10$ and $t \ge 1$.

Case $u, s \ge 2$. Then $T \le \varphi(r) + \varphi(t) + 1 + a(R_1)a(R'') \le 4r + 4t + 1 + rt \le 4d + 1 + d^2/4$, and $4d + 1 + d^2/4 \le \binom{d}{2} + d - 2$ is equivalent to $d^2 - 14d - 12 \ge 0$, which holds for $d \ge 15$. For $10 \le d \le 14$ we instead evaluate (5) for all root lattices of rank d (88, 128, 193, 276, 403 lattices) and find $\overline{\mathcal{N}}(R) \le \binom{d+1}{2} - 2$ for all $R \ne A_d$; Table 2 lists the largest values. (The evaluation was in fact carried out for all $d \le 22$, with the same outcome, and it takes a few seconds; the script is part of the ancillary material.)

This completes the induction. □

d	#lattices	$\binom{d+1}{2}$	D_d	$A_{d-1} \oplus A_1$	$D_{d-1} \oplus A_1$	other largest $\overline{\mathcal{N}}$
9	57	45	37	36	30	$E_8 \oplus A_1$: 45, $E_7 \oplus A_2$: 36
10	88	55	46	45	38	$E_8 \oplus A_2$, $E_8 \oplus A_1 \oplus A_1$: 45
11	128	66	56	55	47	$E_8 \oplus A_3$, $A_9 \oplus A_2$, $E_8 \oplus A_2 \oplus A_1$: 45
12	193	78	67	66	57	$A_{10} \oplus A_2$: 55
13	276	91	79	78	68	$A_{11} \oplus A_2$: 66
14	403	105	92	91	80	$A_{12} \oplus A_2$: 78
15	570	120	106	105	93	$A_{13} \oplus A_2$: 91
16	815	136	121	120	107	$A_{14} \oplus A_2$: 105
17	1137	153	137	136	122	$A_{15} \oplus A_2$: 120
18	1599	171	154	153	138	$A_{16} \oplus A_2$: 136
19	2207	190	172	171	155	$A_{17} \oplus A_2$: 153
20	3058	210	191	190	173	$A_{18} \oplus A_2$: 171
21	4177	231	211	210	192	$A_{19} \oplus A_2$: 190
22	5713	253	232	231	212	$A_{20} \oplus A_2$: 210

TABLE 2. The bound $\overline{\mathcal{N}}$ of (5) for the root lattices of rank d other than A_d: the values for D_d, $A_{d-1} \oplus A_1$, $D_{d-1} \oplus A_1$ (the first two are exact by Theorems 3.2, 4.2, the third is exact by the construction in the remark after Lemma 5.1) and the next largest value. For $d \ge 10$ all values are at most $\binom{d+1}{2} - 2$.

Corollary 6.2. *For* $10 \le d \le 22$ *the three largest values of* $\mathcal{N}(R)$ *over the root lattices* R *of rank* d *are* $\binom{d+1}{2}$ *(only for* A_d*),* $1 + \binom{d}{2}$ *(only for* D_d*) and* $\binom{d}{2}$ *(attained by* $A_{d-1} \oplus A_1$*).*

Proof. $\mathcal{N}(A_d)$ and $\mathcal{N}(D_d)$ are Theorems 3.2 and 4.2, $\mathcal{N}(A_{d-1} \oplus A_1) \ge \mathcal{N}(A_{d-1}) = \binom{d}{2}$ and $\overline{\mathcal{N}}(A_{d-1} \oplus A_1) = \binom{d}{2}$ (the cross term is $\nu_0(A_{d-1}) + \nu_0(A_1) + 1 = \lfloor d^2/4 \rfloor + 2$ or smaller); every other root lattice of rank d has $\overline{\mathcal{N}}(R) \le \binom{d}{2}$ by the evaluation summarised in Table 2. □

Corollary 6.2 answers, in the range $10 \le d \le 22$, the question of [21, Remark 8.3]: among connected configurations (Section 7) with squared distances $2, 4$ in $\mathbb{R}^d$ that do not come from A_d,

the largest are the sets $\{0\}\cup\{e_i+e_j\}$ of D_d, with $1+\binom{d}{2}$ points. We expect the same for all $d\ge 10$; the estimates in the proof of Theorem 6.1 give only $\binom{d+1}{2}-2$.

7. From lattices to Euclidean space

We recall the reduction of [21, Sections 3–4], with proofs, in the form needed here. Let $X\subset\mathbb{R}^d$ be a $\{2,4\}$-set, and let $\Gamma(X)$ be the graph on X in which $x\sim y$ when $\|x-y\|^2=2$ (the *short-distance graph*).

Lemma 7.1. *Let $X\subset\mathbb{R}^d$ be a $\{2,4\}$-set with $\Gamma(X)$ connected, and let $p\in X$. Then the differences $x-y$ with $\|x-y\|^2=2$ generate a root lattice $R(X)$ of rank $\dim\operatorname{aff}(X)\le d$, and $X-p\subset R(X)$ is an admissible set. In particular $|X|\le\mathcal{N}(R(X))$.*

Proof. For $x,y,z\in X$ the polarisation identity gives $\langle x-z,y-z\rangle=\frac12(\|x-z\|^2+\|y-z\|^2-\|x-y\|^2)\in\mathbb{Z}$, since all three norms lie in $\{0,2,4\}$ and their sum is even; hence all inner products of differences of points of X are integers, and the group Λ generated by $X-X$ is generated by finitely many vectors of even norm with integral inner products, so it is an even lattice. The subgroup $R(X)$ generated by the short differences is an even lattice generated by norm-2 vectors, hence a root lattice. Since $\Gamma(X)$ is connected, every $x-p$ is a sum of short differences along a path, so $X-p\subset R(X)$, and $R(X)$ spans the same space as $X-X$, of dimension $\dim\operatorname{aff}(X)$. Finally $X-p$ contains 0 and is a $\{2,4\}$-set, i.e. it is admissible. □

Lemma 7.2 ([21, Lemma 4.1])**.** *Let $X\subset\mathbb{R}^d$ be a $\{2,4\}$-set whose short-distance graph has $m\ge 2$ connected components $X_1,\dots,X_m$, let $U_i=\operatorname{span}(X_i-X_i)$ and $a_i=\dim U_i$. Then $U_i\perp U_j$ for $i\ne j$; each X_i lies on a sphere with centre $c_i\in\operatorname{aff}(X_i)$ and radius ρ_i; $c_i-c_j\perp U_k$ for all i,j,k, and $\|c_i-c_j\|^2+\rho_i^2+\rho_j^2=4$ for $i\ne j$; and*

$$\dim\operatorname{aff}(X)=\sum_i a_i+\dim\operatorname{span}\{c_i-c_1:i\le m\},$$

which is at least $\sum_i a_i+1$ if two of the X_i are single points. Moreover $|X_i|\le f(a_i)$, where $f(a)=\max\{\mathcal{N}(R):R$ a root lattice of rank $\le a\}$ and $f(0)=1$.

Proof. Points in different components are at squared distance 4. For $x,x'\in X_i$ and $y,y'\in X_j$ with $i\ne j$, expanding $\|x-y\|^2=\|x-y'\|^2=\|x'-y\|^2=\|x'-y'\|^2=4$ gives $\langle x-x',y-y'\rangle=0$; this is $U_i\perp U_j$. Fix $j\ne i$ and $y\in X_j$: all of X_i lies on the sphere of radius 2 about y, hence on the sphere in $\operatorname{aff}(X_i)$ with centre the orthogonal projection c_i of y to $\operatorname{aff}(X_i)$; as X_i spans $\operatorname{aff}(X_i)$ this sphere is unique, so c_i and ρ_i do not depend on y or j, and $y-c_i\perp U_i$ for all $y\notin X_i$. For $i\ne j$ and $y\in X_j$, $c_i-c_j=(c_i-y)+(y-c_j)$ with $c_i-y\perp U_i$ and $y-c_j\in U_j\perp U_i$, so $c_i-c_j\perp U_i$, and symmetrically $\perp U_j$; for $k\notin\{i,j\}$ write $c_i-c_j=(c_i-c_k)-(c_j-c_k)$. For $x\in X_i$, $y\in X_j$ the three vectors $x-c_i\in U_i$, $c_i-c_j\in(U_i\oplus U_j)^\perp$, $c_j-y\in U_j$ are mutually orthogonal, whence $4=\|x-y\|^2=\rho_i^2+\|c_i-c_j\|^2+\rho_j^2$. The set $\operatorname{aff}(X)-c_1$ is spanned by the orthogonal subspaces $U_1,\dots,U_m$ and $\operatorname{span}\{c_i-c_1\}$; if $X_i=\{p\}$, $X_j=\{q\}$ are single points then $c_i=p\ne q=c_j$. Finally, X_i is a $\{2,4\}$-set (or a single point) with connected short-distance graph in the a_i-dimensional space $\operatorname{aff}(X_i)$, so $|X_i|\le\mathcal{N}(R(X_i))\le f(a_i)$ by Lemma 7.1. □

Put $\tilde f(0)=1$, $\tilde f(1)=2$, $\tilde f(a)=\binom{a+2}{2}$ for $2\le a\le 8$ and $\tilde f(a)=\binom{a+1}{2}$ for $a\ge 9$. Then $f(a)\le\tilde f(a)$ for all $a\ge 0$: for $a\ge 9$ by Theorem 6.1, for $2\le a\le 8$ by (2), and $f(1)=2$, $f(0)=1$ trivially. (The exact values are $f(0),\dots,f(8)=1,2,4,6,10,16,27,29,45$ [21, Table 1], but we do not need them.)

Lemma 7.3. *Let $d\ge 10$, $m\ge 2$, and let $a_1\ge a_2\ge\dots\ge a_m\ge 0$ be integers with $\sum_i a_i\le d$; let s be the number of indices with $a_i=0$, and assume $\sum_i a_i\le d-s+1$ if $s\ge 2$. If $(a_1,\dots,a_m)\ne(d,0)$, then $\sum_i\tilde f(a_i)\le\binom{d+1}{2}-1$.*

Proof. Let $b_1, \dots, b_q$ be the positive a_i and $A = \sum_j b_j$, so $\sum_i \tilde{f}(a_i) = \sum_j \tilde{f}(b_j) + s$ and $q + s = m$. For $b \geq 1$ we have $\tilde{f}(b) \leq \binom{b+1}{2} + b + 1$, and for $1 \leq b \leq 8$ also $\tilde{f}(b) \leq \frac{45}{8} b$.

Case $a_1 \geq 9$. Let $B = A - a_1$ be the sum of the other positive parts. If $B \geq 1$ then, using $\sum_{j \geq 2} \binom{b_j+1}{2} \leq \binom{B+1}{2}$,

$$\sum_j \tilde{f}(b_j) \leq \binom{a_1+1}{2} + \binom{B+1}{2} + 2B = \binom{a_1+B+1}{2} - (a_1-2)B \leq \binom{A+1}{2} - 7B.$$

If $s \leq 1$ this is at most $\binom{d+1}{2} - 7$, and $\sum_i \tilde{f}(a_i) \leq \binom{d+1}{2} - 6$. If $s \geq 2$ then $A \leq d - s + 1$, so $\binom{A+1}{2} \leq \binom{d+1}{2} - \frac{s-1}{2}(2d - s + 2) \leq \binom{d+1}{2} - 6(s-1)$, and $\sum_i \tilde{f}(a_i) \leq \binom{d+1}{2} - 6(s-1) - 7 + s < \binom{d+1}{2} - 1$. If $B = 0$ then $m - 1 = s \geq 1$; for $s = 1$ we have $a_1 \leq d - 1$ (as $(a_1, 0) \neq (d, 0)$) and $\sum_i \tilde{f}(a_i) \leq \binom{d}{2} + 1 \leq \binom{d+1}{2} - 1$; for $s \geq 2$, $a_1 \leq d - s + 1 \leq d - 1$ and $\sum_i \tilde{f}(a_i) \leq \binom{d-s+2}{2} + s \leq \binom{d}{2} + 2 \leq \binom{d+1}{2} - 1$.

Case $a_1 \leq 8$. If $q = 0$ then $\sum_i \tilde{f}(a_i) = m \leq d + 1$. Otherwise $\sum_j \tilde{f}(b_j) \leq \frac{45}{8} A$. If $s \leq 1$ then $\sum_i \tilde{f}(a_i) \leq \frac{45}{8} d + 1$, which is at most $\binom{d+1}{2} - 1$ for $d \geq 11$; for $d = 10$ the partitions of $A \leq 10$ into parts at most 8 give $\sum_j \tilde{f}(b_j) \leq \tilde{f}(8) + \tilde{f}(2) = 51$, so the sum is at most 52. If $s \geq 2$ then $A \leq d - s + 1$ and $\sum_i \tilde{f}(a_i) \leq \frac{45}{8}(d+1) - \frac{37}{8} s \leq \frac{45}{8}(d+1) - \frac{37}{4} \leq \binom{d+1}{2} - 1$ for $d \geq 10$. □

Proof of Theorem 1.1. Let $d \geq 10$ and let $X \subset \mathbb{R}^d$ be a $\{2, 4\}$-set with $|X| \geq \binom{d+1}{2}$.

If $\Gamma(X)$ is connected, then by Lemma 7.1 and Theorem 6.1, $|X| \leq \mathcal{N}(R(X)) \leq \binom{d+1}{2}$, with equality only if $R(X) \cong A_d$; and then $X - p$ is an admissible subset of A_d of size $\binom{d+1}{2}$, hence congruent to $\mathcal{J}(d+1, 2)$ by Theorem 3.2. Conversely $\mathcal{J}(d+1, 2)$ is a $\{2, 4\}$-set with $\binom{d+1}{2}$ points spanning a d-dimensional space.

Suppose $\Gamma(X)$ is disconnected, with components $X_1, \dots, X_m$ numbered so that $a_1 \geq \cdots \geq a_m$, where $a_i = \dim \operatorname{aff}(X_i)$. By Lemma 7.2, $\sum_i a_i \leq d$, and if $s \geq 2$ of the components are single points then these points are pairwise at distance 2, hence affinely independent, so that $\sum_i a_i + s - 1 \leq \dim \operatorname{aff}(X) \leq d$. Moreover $|X| = \sum_i |X_i| \leq \sum_i f(a_i) \leq \sum_i \tilde{f}(a_i)$. By Lemma 7.3 and $|X| \geq \binom{d+1}{2}$ we get $m = 2$, $a_1 = d$, $a_2 = 0$: $X = X_1 \cup \{p\}$ with X_1 connected, $\dim \operatorname{aff}(X_1) = d$ and $|X_1| \geq \binom{d+1}{2} - 1$.

Now Lemma 7.2 gives $d = \dim \operatorname{aff}(X) = d + \dim \operatorname{span}\{c_2 - c_1\}$, so $c_1 = c_2 = p$, and then $\rho_1^2 = 4 - \|c_1 - c_2\|^2 - \rho_2^2 = 4$: the set X_1 lies on the sphere of squared radius 4 about p. On the other hand $X_1 - p'$ ($p' \in X_1$) is an admissible subset of the root lattice $R(X_1)$ of rank d with at least $\binom{d+1}{2} - 1$ points, so $R(X_1) \cong A_d$ by Theorem 6.1, and X_1 is congruent to $\mathcal{J}(d+1, 2)$ or to $\mathcal{J}(d+1, 2)$ minus a point by Theorem 3.2. These sets span $\mathbb{R}^d$ and lie on a sphere of squared radius $2 - 4/(d+1) < 2$, hence on no sphere of squared radius 4. This contradiction shows that $\Gamma(X)$ is connected, and the theorem follows. □

Proof of Theorem 1.2. Let $10 \leq d \leq 21$ and let $X \subset \mathbb{R}^d$ be a two-distance set with $|X| \geq \binom{d+1}{2}$, with distances $a < b$. Since $\binom{d+1}{2} > 2d + 1$, the theorem of Larman, Rogers and Seidel in Neumaier's form ([16, 19]; [21, Theorem 2.1]) gives $a^2/b^2 = (k-1)/k$ for an integer k with $2 \leq k \leq \frac{1}{2} + \sqrt{d/2} < 4$. If $k = 2$, then after scaling X is a $\{2, 4\}$-set and Theorem 1.1 gives $|X| \leq \binom{d+1}{2}$ with equality only for $\mathcal{J}(d+1, 2)$. If $k = 3$, then [7, Theorem 3.1] (see also [7, Table 2]) gives

$$|X| \leq \frac{24(d+1)}{24 - d} + 1 < \binom{d+1}{2} \qquad (13 \leq d \leq 21),$$

the largest value of the left-hand side being 177 at $d = 21$; for $d \leq 12$ the case $k = 3$ does not occur. Hence $g(d) = \binom{d+1}{2}$, attained only by sets similar to $\mathcal{J}(d+1, 2)$. □

8. Computations and verification

All computations are small and were carried out with two independent tools: a branch-and-bound maximum-clique program written in C for this paper (bitset representation, greedy-colouring bound in the style of Tomita and Seki [23]), and the program `cliquer` of Niskanen and Östergård [20]. The graphs are generated by short Python scripts from the coordinates of Section 2 (with all vectors doubled so that coordinates are integers). Code, graphs and logs accompany the paper.

8.1. **The invariants of E_6, E_7, E_8.** The values of $a(R)$ and $\nu_0(R)$ in Table 1 are the clique numbers of two graphs on the 72, 126, 240 roots. Both programs return $a(E_6) = 5$, $a(E_7) = 7$, $a(E_8) = 8$ and $\nu_0(E_6) = 16$, $\nu_0(E_7) = 27$; for $\nu_0(E_8)$ the C program returns 36 ($5.6 \cdot 10^7$ search nodes), and `cliquer` confirms 35 for the clique number of the subgraph induced on the 182 roots at inner product 0 or 1 with a fixed root (every admissible set is $W(E_8)$-equivalent to one containing that root), i.e. $\nu_0(E_8) = 1 + 35 = 36$. For $k \geq 1$ the values ν_k were computed in two ways: by the method described in the proof of Proposition 5.3 (fixing one root or one 60°-pair), and by a C program that enumerates every 60°-clique K of R_2 and computes, for each, the clique number of the admissibility graph on the common neighbourhood $\{y : \langle y, w\rangle = 1 \ \forall w \in K\}$. For E_8 the second program visits 240, 6720, 60480, 241920, 483840, 483840, 207360, 17280 cliques of sizes $1, \dots, 8$ and returns $\nu_1, \dots, \nu_8 = 29, 29, 19, 14, 11, 9, 8, 8$, in agreement with the first; for E_6 and E_7 the same agreement holds. The same enumeration for A_4, A_5, A_6 and $D_4, \dots, D_8$ reproduces the formulas of Propositions 3.4 and 4.4, and gives $\nu_k(D_n) = 2n - k - 2$ for $3 \leq k \leq n - 1$ in this range.

8.2. **Positive controls.** As a check of the graph construction and of the solvers, we recomputed $\mathcal{N}(R) = 1 + \omega(G(R))$ by a plain clique search (no symmetry reduction) for all 39 root lattices of rank 8 and all 57 root lattices of rank 9, and for several other lattices. The results agree with [21, Tables 1 and 2] in every case; in particular

$$\mathcal{N}(E_6) = 27, \quad \mathcal{N}(E_7) = 29, \quad \mathcal{N}(E_8) = 45, \quad \mathcal{N}(E_8 \oplus A_1) = 45, \quad \mathcal{N}(A_9) = 45, \quad \mathcal{N}(D_9) = 37,$$

$\mathcal{N}(A_n) = \binom{n+1}{2}$ for $2 \leq n \leq 9$, and $\mathcal{N}(D_n) = 8, 16, 17, 22, 29, 37, 46$ for $n = 4, \dots, 10$ (the values $n \leq 6$ are used in Theorem 4.2; the values $7 \leq n \leq 10$ agree with Theorem 4.2, and $\mathcal{N}(D_{10}) = 46$ is also in [21, Table 1]). The largest of these searches, $G(E_8)$ with 2400 vertices and $G(E_8 \oplus A_1)$ with 2882 vertices, took about three and four minutes on one core ($7 \cdot 10^7$ and $9 \cdot 10^7$ nodes); $G(D_{10})$ with 3560 vertices took two minutes.

8.3. **The bound $\overline{\mathcal{N}}$.** The recursive bound (5) was evaluated for all root lattices of rank at most 22 (with the minimum over all decompositions into two sub-sums when the number of components is at most 7, and over the decompositions that split off one component otherwise). Table 2 summarises the outcome; the full lists are in the ancillary files. Comparing with the exact values of the previous paragraph, $\overline{\mathcal{N}}(R) = \mathcal{N}(R)$ for 25 of the 39 lattices of rank 8 and for 34 of the 57 lattices of rank 9, and $\overline{\mathcal{N}}(R) - \mathcal{N}(R) \leq 10$ throughout; the largest gaps occur for lattices with a D_n or E_7 summand and a small complement (for instance $\overline{\mathcal{N}}(D_6 \oplus A_1^{\oplus 3}) = 28$ against $\mathcal{N} = 19$), where the term $\nu_0(L') + \nu_0(L'') + 1$ of Lemma 5.1 counts admissible root sets of the two summands that cannot be combined. As a consistency check, $\overline{\mathcal{N}}(R) \geq \mathcal{N}(R)$ held in all 96 cases, as it must.

9. Concluding remarks

Dimensions 22 and beyond. Theorem 1.1 holds in every dimension $d \geq 10$, but it determines $g(d)$ only where the other distance ratios are excluded. For $d = 22$ the bound of [7] for $k = 3$

is 277, and there is a spherical two-distance set of 275 points in $\mathbb{R}^{22}$ with squared distances in the ratio $2:3$, coming from the McLaughlin graph [18, 5]; so $g(22) \geq 275 > 253 = \binom{23}{2}$. For $d = 23$ the 276 equiangular lines in $\mathbb{R}^{23}$ give a spherical two-distance set of 276 points, and Ge, Koolen and Munemasa [13] found a two-distance set of 277 points with distances 2 and $\sqrt{6}$; so $g(23) \geq 277 > 276 = \binom{24}{2}$. In both dimensions the extremal sets have $k = 3$, and Theorem 1.1 only says that they are not of the type $k = 2$. For $d \geq 24$ the theorem of Chen and Yu gives no finite bound for $k = 3$ (their condition $(2k-1)^2 > d+1$ fails), and the determination of $g(d)$ requires an analysis of the sets with squared-distance ratio $2:3$. It would be interesting to know whether such sets admit a lattice description similar to the one used here: their differences of norm 2 and 3 (in the natural scaling) have half-integral inner products, so that $\sqrt{2}$ times the difference lattice is an integral lattice generated by vectors of norm 4 and 6, but no analogue of Witt's classification is available.

Largest configurations from other lattices. Corollary 6.2 shows that for $10 \leq d \leq 22$ the largest $\{2,4\}$-sets in $\mathbb{R}^d$ with connected short-distance graph that do not come from A_d are the $(1 + \binom{d}{2})$-point sets $\{0\} \cup \{e_i + e_j\}$ of D_d; the sets $\mathcal{J}(d,2)$ with a point added at distance $\sqrt{2}$ from all of $\mathcal{J}(d,2)$ are the same sets. We expect this for all $d \geq 10$, and it would follow from a sharpening of the estimates in the proof of Theorem 6.1 by $d-3$. Among the $\{2,4\}$-subsets of A_d itself, Theorem 2.4 shows that every family of weight w with $3 \leq w \leq d-2$ has at most $\max\{\binom{d-w+3}{2}, 1 + w(d+1-w), \binom{w+2}{2}\}$ members; for $w = 3$ this is $\binom{d}{2}$, attained by the 3-sets through a fixed point, which as a point set is again $\mathcal{J}(d,2)$.

Two-distance sets with ratio $\sqrt{2}$ on the sphere. For spherical two-distance sets the maximum $d(d+1)/2$ was established by Musin [18] for $7 \leq d \leq 39$, $d \neq 22, 23$, by Barg and Yu [5] for $d = 23$ and $40 \leq d \leq 93$, $d \neq 46, 78$, and by Glazyrin and Yu [14] for all $d \geq 7$ outside the dimensions $d = (2k+1)^2 - 3$, by the polynomial method and without restriction on the ratio. Theorem 1.1 is a statement about a fixed ratio but without the sphericity assumption; its proof is combinatorial and does not use the polynomial method except through Blokhuis' bound for the lattices of rank at most 8, which enters only through the value 45 in (6) and in $\tilde{f}$. The two results overlap for spherical sets with ratio $\sqrt{2}$.

The method. The lattices A_n and D_n account for the size of the graphs $G(R)$ in [21]; the observation that their $\{2,4\}$-sets are, in disguise, uniform set systems with two intersection sizes and binary codes with two distances makes the extremal theory of these objects available. The slicing lemma is elementary but effective: it shows that the interaction between orthogonal summands is governed by the small quantities a and ν_k, and reduces the whole family of root lattices, whose number grows like the number of partitions of d, to a short table. We expect the same three ingredients to apply to other lattice-theoretic extremal problems in which the relevant vectors lie in a root lattice, for instance to sets with squared distances $\{2,4,6\}$, where however the ratios are no longer forced by a Larman–Rogers–Seidel type theorem.

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